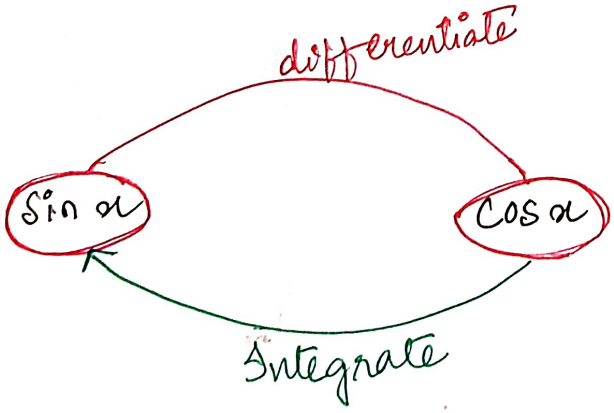
-: 12th BOARD HACKER JUGAAD NOTES :-

INTEGRALS

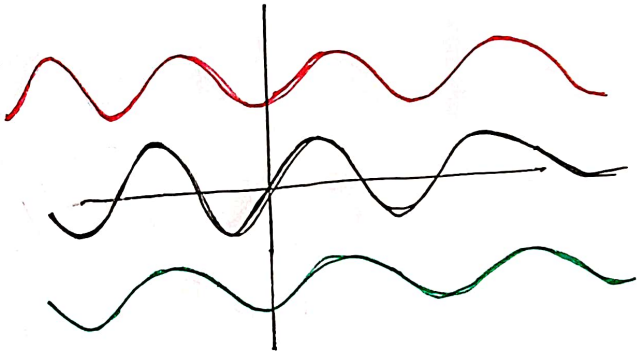
1. Integration is the reverse process of differentiation.
2. Integration means Summation.
3. Geometrically the statement represents a family of curves the different values of C correspond to different members of this family.
4. The concept of integration originated during the course of finding area of plane figure.



$\int \cos x\,dx = \sin x + C$ → constant of integration.

- $\int$: Symbol represents Integration
- $\cos x\,dx$: Integrand
- $\sin x + C$: Integral
- x → valuable of Integration.

$$\int \cos x\,dx = \sin x + C$$
$$= \sin x + 2$$
$$\sin x - 2$$



INTEGRALS

- Indefinite Integrals
- Definite Integrals

$$\int \cos x\,dx = \sin x + C$$

$$\int_{\pi/4}^{\pi/2} \cos x\,dx$$

$\pi/2$: upper limit; $\pi/4$: lower limit

Derivatives

i) $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$

Particularly, we note that

$\frac{d}{dx}(x) = 1$

ii) $\frac{d}{dx}(\sin x) = \cos x$

iii) $\frac{d}{dx}(-\cos x) = \sin x$

iv) $\frac{d}{dx}(\tan x) = \sec^2 x$

v) $\frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x$

vi) $\frac{d}{dx}(\sec x) = \sec x \tan x$

vii) $\frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x$

viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

ix) $\frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

xi) $\frac{d}{dx}(-\cot x) = \frac{1}{1+x^2}$

xii) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$

xiii) $\frac{d}{dx}(-\operatorname{cosec}^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$

xiv) $\frac{d}{dx}(e^x) = e^x$

xv) $\frac{d}{dx}(\log|x|) = \frac{1}{x}$

xvi) $\frac{d}{dx}\left(\frac{a^x}{\log a}\right) = a^x$

Integrals (Anti Derivatives)

i) $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$

ii) $\int dx = x + c$

iii) $\int \cos x\,dx = \sin x + c$

iv) $\int \sin x\,dx = -\cos x + c$

v) $\int \sec^2 x\,dx = \tan x + c$

vi) $\int \operatorname{cosec}^2 x\,dx = -\cot x + c$

vii) $\int \sec x \tan x\,dx = \sec x + c$

viii) $\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + c$

ix) $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$

x) $\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + c$

xi) $\int \frac{dx}{1+x^2} = \tan^{-1} x + c$

xii) $\int \frac{dx}{1+x^2} = -\cot^{-1} x + c$

xiii) $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c$

xiv) $\int \frac{dx}{x\sqrt{x^2-1}} = -\operatorname{cosec}^{-1} x + c$

xv) $\int e^x\,dx = e^x + c$

xvi) $\int \frac{1}{x}\,dx = \log|x| + c$

xvii) $\int a^x\,dx = \frac{a^x}{\log a} + c$

Eg: $\int x^7 dx = \frac{x^{7+1}}{7+1} + C = \frac{x^8}{8} + C$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

eg. $\int 1 dx = \int x^0 dx$

$= \frac{x^{0+1}}{0+1} + C$

$= x + C$

eg. $\int 2^x dx = \frac{2^x}{\log 2} + C$

$$\int a^x dx = \frac{a^x}{\log a}$$

eg. $\int x^{\frac{1}{2}} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$

$= \frac{x^{3/2}}{3/2} + C$

$= \frac{2}{3} x^{3/2} + C$

Properties :-

1. $\int [f_1(x) + f_2(x) + f_3(x)] dx = \int f_1(x) dx + \int f_2(x) dx + \int f_3(x) dx$

2. $\int [K_1 f_1(x) + K_2 f_2(x) + K_3 f_3(x)] dx$
$= K_1 \int f_1(x) dx + K_2 \int f_2(x) dx + K_3 \int f_3(x) dx$

K_1, K_2, K_3 Constant

eg. $\int (3\cos x + 7x^2) dx$

$\Rightarrow \int 3\cos x dx + \int 7x^2 dx$

$\Rightarrow 3\int \cos x dx + 7\int x^2 dx$

$\Rightarrow 3 \sin x + 7 \times \frac{x^{2+1}}{2+1} + C$

$\Rightarrow 3 \sin x + \frac{7}{3} x^3 + C$ Ans.

Methods of Integration

1. Integration by substitution
2. Integration using partial functions.
3. Integration by parts.

INTEGRATION BY SUBSTITUTION

$\int f(x) f'(x) dx$ $\quad$ $\int \text{cosec}^2(-8x+7) dx$

$f(x) = t$ $\quad$ $-8x+7 = t$

CHAIN RULE

Argument should be linear $\Rightarrow$ Power of the veriable should be 1

(1) $\int \cos(8x+9) dx = \frac{\sin(8x+9)}{8} + c$

② $\int \sec^2(7x+3)\,dx = \frac{\tan(7x+3)}{7} + c$

③ $\int \sin(-8x+7)\,dx = \frac{-\cos(-8x+7)}{-8} + c$

④ $\int e^{5x-4}\,dx = \frac{e^{5x-4}}{5} + c$

⑤ $\int 7^{2x+3}\,dx = \frac{7^{2x+3}}{\log 7 \times 2} + c$

⑥ $\int \frac{1}{3x+2}\,dx = \frac{\log|3x+2|}{3} + c$

⑦ $\int \cos(x^2+7)\,dx = X$

(② Not linear)

INTEGRATION USING TRIGO-NOMETRIC IDENTITIES

① $\cos 2x = 1 - 2\sin^2 x$

$2\sin^2 x = 1 - \cos 2x$

$$\sin^2 x = \frac{1-\cos 2x}{2}$$

② $\cos 2x = 2\cos^2 x - 1$

$2\cos^2 x = 1 + \cos 2x$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

③
$2\sin A\cos B = \sin(A+B) + \sin(A-B)$
$2\cos A\sin B = \sin(A+B) - \sin(A-B)$
$2\cos A\cos B = \cos(A+B) + \cos(A-B)$
$2\sin A\sin B = \cos(A-B) - \cos(A+B)$

Formulas

1. $\int \tan x\,dx = \log|\sec x| + c$
2. $\int \cot x\,dx = \log|\sin x| + c$
3. $\int \sec x\,dx = \log|\sec x + \tan x| + c$
4. $\int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + c$

INTEGRAL FORM

Form IV : $\int \sin^m x \cos^n x\,dx$

$\int \sin^3 x \cos^2 x\,dx$

$\Rightarrow \int \sin^2 x \cos^2 x \times \sin x\,dx$

$\Rightarrow \int \underbrace{(1-\cos^2 x)\cos^2 x}_{(1-t^2)t^2} \times \underbrace{\sin x\,dx}_{-dt}$

$\Rightarrow \int (1-t^2)t^2 \times -dt$

$\cos x = t$

$-\sin x\,dx = dt$

$\sin x\,dx = -dt$

$-\int t^2(1-t^2)\,dt$

$-\int (t^2 - t^4)\,dt$

$\int (t^4 - t^2)\,dt$

$\frac{t^5}{5} - \frac{t^3}{3} + c$

$\frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + c$ Ans.

INTEGRAL FORM

Form : $\int \frac{dx}{\text{quadratic}}$

1. $\int \frac{1}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$

eg $\int \frac{1}{x^2-9}\,dx = \int \frac{1}{(x)^2-(3)^2}\,dx$

$= \frac{1}{2\times 3}\log\left|\frac{x-3}{x+3}\right| + c$

$= \frac{1}{6}\log\left|\frac{x-3}{x+3}\right| + c$ Ans.

2. $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

eg $\int \frac{1}{a^2-x^2} = \int \frac{1}{(3)^2-(x^2)}\,dx$

$= \frac{1}{2\times 3}\log\left|\frac{3+x}{3-x}\right| + c$

$= \frac{1}{6}\log\left|\frac{3+x}{3-x}\right| + c$ Ans.

3. $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

eg $\int \frac{1}{x^2+a}\,dx$

$\int \frac{1}{(x)^2+(2)^2}\,dx = \frac{1}{2}\tan^{-1}\frac{x}{2} + c$

$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

FORMULA

$\int \frac{1}{\sqrt{\text{quadratic}}}$

$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$

eg $\int \frac{1}{\sqrt{9-x^2}}\,dx = \int \frac{1}{\sqrt{(3)^2-(x)^2}}\,dx$

$= \sin^{-1}\frac{x}{3} + c$ Ans

$\int \frac{dx}{\sqrt{a^2+x^2}} = \log\left|x+\sqrt{x^2+a^2}\right| + c$

eg. $\int \frac{1}{\sqrt{x^2+9}}\,dx$

$= \int \frac{1}{\sqrt{(x)^2+(3)^2}}\,dx = \log\left|x+\sqrt{x^2+3^2}\right| + c$

$= \log\left|x+\sqrt{x^2+9}\right| + c$ Ans

$\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right| + c$

eg. $\int \frac{1}{\sqrt{x^2-64}} =$

$= \int \frac{1}{\sqrt{(x)^2-(8)^2}} = \log\left|x+\sqrt{x^2-(8)^2}\right| + c$

$= \log\left|x+\sqrt{x^2-64}\right| + c$ Ans

INTEGRAL FORM

$\int \frac{(px+q)\,dx}{ax^2+bx+c} = \frac{\text{linear}}{\text{quadratic}}$

$\int \frac{(px+q)\,dx}{\sqrt{ax^2+bx+c}} = \frac{\text{linear}}{\sqrt{\text{quadratic}}}$

$\text{linear} = \frac{Ad(\text{quadratic})}{dx} + B$

INTEGRATION BY PARTS

If u and v are two functions of x, then

$$\int uv\,dx = u\left(\int v\,dx\right) - \int\left\{\frac{du}{dx}\int v\,dx\right\}dx$$

$$\int F\cdot S\,dx = F\int S\,dx - \int\left(\frac{df}{dx}\times\int S\,dx\right)dx$$

I L A T E

- I → Inverse Trigonometric function
- L → Logarithmic function
- A → Algebraic
- T → Trigonometric
- E → exponential

$\int x\log x\,dx$: x → S, $\log x$ → F

$\int (\sin^{-1}x)\,e^x\,dx$: $\sin^{-1}x$ → F, e^x → S

INTEGRAL FORM

$$\int e^x(f(x)+f'(x))\,dx = e^x f(x)+c$$

$$\int e^x[f(x)+f'(x)]\,dx = e^x f(x)+c$$

Ex. $\int e^x[\underbrace{\sin x}_{f(x)}+\underbrace{\cos x}_{f'(x)}]\,dx = e^x f(x)+c = e^x\sin x+c$ Ans

INTEGRATION BY PARTIAL FRACTIONS

$$f(x) = \frac{\text{Numerator}}{\text{Denominator}} = \frac{\text{Polynomial}_1}{\text{Polynomial}_2}$$

Degree of Numerator < Degree of Denominator

Type-01

$$\frac{\text{Numerator}}{(x-a)(x-b)(x-c)} = \frac{A}{(x-a)}+\frac{B}{(x-b)}+\frac{C}{(x-c)}$$

Type-02

$$\frac{\text{Numerator}}{(x-a)^2(x-b)^3(x-c)} = \frac{A}{(x-a)}+\frac{B}{(x-a)^2}+\frac{C}{(x-b)}+\frac{D}{(x-b)^2}+\frac{E}{(x-b)^3}+\frac{F}{(x-c)}$$

Type-03

$$\frac{\text{Numerator}}{(x-a)(x^2+b)} = \frac{A}{(x-a)}+\frac{Bx+C}{(x^2+b)}$$

INTEGRAL FORM

$$\int \sqrt{\text{quadratic}}\, dx$$

1. $$\int \sqrt{a^2+x^2}\, dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\log\left|x+\sqrt{a^2+x^2}\right| + c$$

2. $$\int \sqrt{x^2-a^2}\, dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + c$$

3. $$\int \sqrt{a^2-x^2}\, dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{1}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right| + c$$

$$\int \frac{1}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right| + c$$

$$\int \frac{1}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$$

Qs. $\int \sqrt{x^2+25}\, dx$

$$= \int \sqrt{(x)^2+(5)^2}\, dx = \frac{x}{2}\sqrt{x^2+25} + \frac{(5)^2}{2}\log\left|x+\sqrt{x^2+25}\right| + c \quad \text{Ans.}$$

Q. $\int \sqrt{x^2-25}$

$$= \int \sqrt{(x)^2-(5)^2} = \frac{x}{2}\sqrt{x^2-25} - \frac{(5)^2}{2}\log\left|x+\sqrt{x^2-25}\right| + c \quad \text{Ans.}$$

Q. $\int \sqrt{25-x^2}\, dx$

$$= \int \sqrt{(5)^2-(x)^2}$$

$$= \frac{x}{2}\sqrt{25-x^2} + \frac{(5)^2}{2}\sin^{-1}\frac{x}{5} + c \quad \text{Ans.}$$

DEFINITE INTEGRAL

A definite integral has a unique value. A definite integral is denoted by $\int_a^b f(x)\,dx$, where a is called the lower limit of the integral & b is called the upper limit of the integral.

1. $$\int_a^b f(x)\,dx = F(b) - F(a)$$

where, $\int f(x)\,dx = F(x) + c$

$$\overset{\text{Upper limit}}{\overset{\leftarrow b}{\int_{\downarrow a}}} f(x)\,dx = F(b) - F(a)$$

$\downarrow$ Lower limit

Q. $$\int_{\pi/4}^{\pi/3} \cos x\, dx = \left[\sin x\right]_{\pi/4}^{\pi/3}$$

$$= \sin\frac{\pi}{3} - \sin\frac{\pi}{4}$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \quad \text{Ans.}$$

PROPERTY – 01

Change of variable of integration is immaterial so long as limits on integration remain the same.

$$\int_a^b f(x) = \int_a^b f(t)\,dt$$

$$\int_a^b f(x)\,dx = \int_a^b f(y)\,dy$$

$$= \int_a^b f(x)\,dy = \int_a^b f(t)\,dt$$

PROPERTY – 02

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

eg $\int_{\pi/6}^{\pi/3} \cos x\,dx = -\int_{\pi/3}^{\pi/6} \cos x\,dx$

$$[\sin x]_{\pi/6}^{\pi/3} = -[\sin x]_{\pi/3}^{\pi/6}$$

$$\frac{\sqrt{3}}{2} - \frac{1}{2} = -\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{2}$$

PROPERTY – 3

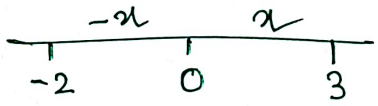
$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

where the point c may lie between a & b.

$$\int_2^5 x^2\,dx = \int_2^3 x^2\,dx + \int_3^5 x^2\,dx$$

$|x| \rightarrow x,\ x \geq 0$

$|x| \rightarrow -x,\ x < 0$



$$\int_{-2}^{3} |x|\,dx = \int_{-2}^{0} -x\,dx + \int_0^3 x\,dx$$

$$= -\left[\frac{x^2}{2}\right]_{-2}^{0} + \left[\frac{x^2}{2}\right]_0^3$$

$$\Rightarrow -\left[0 - \frac{4}{2}\right] + \left[\frac{9}{2} - 0\right]$$

$$\Rightarrow 2 + \frac{9}{2} = \frac{13}{2}$$ Ans.

PROPERTY – 4

V.V. Imp.

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

$x \rightarrow$ U.L + L.L $- x$

$x \rightarrow b + a - x$

$x \rightarrow (a+b-x)$

PROPERTY - 05

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$x \to a+0.x$

$x \to (a-x)$

PROPERTY - 06

$$\int_0^{2a} f(x)\,dx = \int_0^a [f(x)+f(2a-x)]\,dx$$

$$\int_0^{2\pi} \sin x\,dx = \int_0^{\pi} [\sin x + \sin(2\pi - x)]\,dx$$

PROPERTY - 07

If $f(x) = f(2a-x)$, then

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx$$

If $f(x) = -f(2a-x)$, then

$$\int_0^{2a} f(x)\,dx = 0$$

PROPERTY - 8

V.V. Imp.

$$I = \int_{-a}^{a} f(x)\,dx = \int_0^a [f(x)+f(-x)]\,dx$$

$$\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & f(x) \text{ is even function} \\ 0, & f(x) \text{ is odd function.} \end{cases}$$

$$\int_{-a}^{a} f(x)\,dx = \int_0^a [f(x)+f(-x)]\,dx$$

Case I

If $f(x)$ is even function

$$\int_{-a}^{a} f(x)\,dx = \int_0^a f(x)+f(x)$$

$$= \int_0^a f(x)+f(x)$$

$$= \int_0^a f(x)+f(x)$$

$$= \int_0^a 2f(x)\,dx$$

$$= 2\int_0^a f(x)\,dx$$

Case-2

If $f(x)$ is odd function

$f(x) = -f(x)$

$$\int_{-a}^{a} f(x)\,dx = \int_0^a f(x)+f(-x)$$

$$= \int_0^a f(x) - f(x) = 0$$

If $f(x)$ is an odd function.

$\Rightarrow f(-x) = -f(x)$, then $\int_{-a}^{a} f(x)\,dx = 0$